University College London
Department of Physics and Astronomy

Astrophysics Project Report

ExoMol line lists – the rovibrational quadrupole spectrum of the N_2 ground-state

Author:
Jake Clark Brown
(MSci Astrophysics)

Supervisor:
Prof Jonathan Tennyson

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Abstract

This work presents comprehensive molecular line lists for electric quadrupole transitions within the ground electronic state of all three stable isotopologues of molecular nitrogen: $^{14}N_2$, $^{14}N^{15}N$, and $^{15}N_2$. Molecular nitrogen is expected to be abundant in many planetary atmospheres, but its ground-state spectrum is intrinsically weak, making remote detection challenging, though such detections may become increasingly feasible as instrumental sensitivity improves. Existing HITRAN line lists provide accurate data for the strongest transitions but offer limited coverage of highly excited states and do not consistently treat all isotopologues, with $^{15}N_2$ absent entirely, restricting their applicability at elevated temperatures.

Line lists were generated within the ExoMol framework using the Duo nuclear-motion code from a high-accuracy direct-potential-fit potential energy curve and an *ab initio* quadrupole moment function, including Born–Oppenheimer breakdown corrections where required. The calculations are based on the same underlying molecular model as the existing HITRAN data, providing a direct validation reference while extending coverage to higher excitation and all stable isotopologues. All HITRAN transitions for $^{14}N_2$ and $^{14}N^{15}N$ were reproduced, with line positions agreeing extremely well for $^{14}N_2$ and showing small systematic differences for $^{14}N^{15}N$. Transition frequency uncertainties were estimated for all calculated lines, and transition intensities were reproduced to sub-percent accuracy.

To ensure completeness, calculations were extended to high rovibrational excitation ($v, J \leq 80$) and subsequently filtered to remove transitions affected by numerical noise. The final line lists contain 3705, 3070, and 1898 transitions for $^{14}N_2$, $^{14}N^{15}N$, and $^{15}N_2$, respectively, extending to approximately $2 \times 10^4\,\mathrm{cm}^{-1}$ and including hot bands and overtone transitions arising from states with $v \leq 9$. The resulting datasets therefore provide substantially improved completeness at elevated temperature through the inclusion of additional hot-band and high-J transitions, and present the first complete line list for $^{15}N_2$. Simulated spectra and partition functions confirm the physical consistency of the line lists and demonstrate their suitability for modelling nitrogen-rich atmospheres across a wide range of temperatures and for inclusion in the ExoMol database.

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1 Introduction

Molecular spectroscopy provides a powerful means of detecting and characterising gases in planetary atmospheres, the interstellar medium, and laboratory environments. Interpretation of observed spectra relies on molecular line lists, which catalogue the frequencies and intensities of transitions between quantised molecular energy levels [31]. Nitrogen is expected to be a major constituent of many planetary atmospheres, and its ground-state spectrum provides an important tracer of bulk composition across a wide range of temperatures. However, these transitions are intrinsically weak and N_2 remains challenging to detect remotely. Existing line lists provide coverage of the strongest transitions but remain incomplete at higher excitation, limiting their applicability at elevated temperatures [9]. As instrumental sensitivity continues to improve, highly complete and reliable line lists are required to support future observational studies of nitrogen-rich atmospheres.

Nitrogen is chemically stable due to its strong triple bond and is therefore expected to remain abundant across a wide range of physical conditions, from cold protoplanetary environments [20] to hot exoplanet atmospheres [25]. Thermochemical models commonly predict N_2 to be the dominant nitrogen-bearing species in hot Jupiters and in many terrestrial and super-Earth atmospheres [2, 16], meaning its ground-state spectrum probes the dominant atmospheric composition rather than a minor constituent. Early spectroscopic studies of N_2 focused primarily on ultraviolet electronic transitions observed in auroral and upper-atmospheric environments [28, 29]. Such features arise from non-thermal processes in emission or require a far-UV illuminating source in absorption, and are therefore not generally representative of the bulk atmosphere. By contrast, transitions within the ground electronic state arise from rotational and vibrational motion (rovibrational transitions) and lie in the infrared, providing a direct probe of atmospheric composition. However, because N_2 has no permanent dipole moment, these transitions occur via the much weaker electric quadrupole interaction [27], meaning their observation generally requires long path lengths or high column densities together with highly accurate spectroscopic data.

High-quality quadrupole line lists for N_2 are available through the HITRAN database [9], based on modern work that combines laboratory measurements with theoretical calculations to produce accurate molecular models [12, 13]. However, these datasets were developed primarily for terrestrial conditions and therefore provide limited coverage of highly excited states and do not consistently treat all stable isotopologues ($^{14}N_2$, $^{14}N^{15}N$, $^{15}N_2$), restricting their applicability at elevated temperatures such as those found in hot exoplanet atmospheres. The present work builds on the same molecular framework to produce comprehensive line lists for all three isotopologues, extending coverage to higher excitation as part of the ExoMol project [31].

The following sections outline the theoretical background and computational methodology used to generate and validate the line lists, followed by comparison with existing HITRAN data and assessment of completeness at high excitation.

2 Theory & Existing Work

A molecular line list is a catalogue of all radiative transitions between quantised energy levels of a molecule, providing both the transition frequencies and their intensities. The ExoMol project aims to produce comprehensive molecular line lists that serve as fundamental input for modelling and interpreting molecular spectra, particularly for molecules expected to be important in the atmospheres of exoplanets and other hot astronomical environments (such as brown dwarfs and cool stars), where highly complete datasets are required for reliable spectroscopic modelling [31]. Since molecules possess extremely large numbers of rovibrational transitions, it is generally impossible to measure every spectral line experimentally. Instead, limited laboratory measurements are combined with theoretical calculations to generate complete datasets. In practice, this involves solving the nuclear-motion Schrödinger equation within the Born–Oppenheimer approximation, using an accurate potential energy curve (PEC) to determine energy levels and a corresponding molecular moment function (e.g. dipole or quadrupole) to determine transition intensities.

2.1 Transition frequencies

Molecular spectra can arise from electronic, vibrational, or rotational motion. This work only considers transitions within the ground electronic state of N_2, since electronic transitions lie in the ultraviolet and involve energy levels that are not populated under thermal conditions. The molecule is not rigid; even within a single electronic state, the nuclei can rotate and vibrate.

These motions are quantised, giving discrete rotational and vibrational energy levels labelled by the quantum numbers J (rotational) and v (vibrational). Transitions between levels with different v and J are known as *rovibrational transitions*, whose frequencies are determined by solving the nuclear-motion Schrödinger equation.

2.1.1 Nuclear-motion equation and the Born–Oppenheimer approximation

The nuclear-motion Schrödinger equation for a diatomic molecule, such as N_2, is,

$$\hat{H}\Psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{R}_A, \mathbf{R}_B) = E\Psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{R}_A, \mathbf{R}_B), \tag{1}$$

where $\mathbf{r}_1, \mathbf{r}_2$ and $\mathbf{R}_A, \mathbf{R}_B$ are the electronic and nuclear coordinates, respectively, and the Hamiltonian $\hat{H}$ describes the motion of both. This makes a direct solution impractical, so the Born–Oppenheimer approximation is used to separate electronic and nuclear motion.

Electrons are much lighter than nuclei and therefore move much more rapidly, responding almost instantaneously to changes in nuclear position. The nuclei can thus be treated as fixed when solving the electronic problem, allowing the molecular wavefunction to be approximated as

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, R, \theta, \phi) \approx \psi_{\mathrm{el}}(\mathbf{r}_1, \mathbf{r}_2; R)\, \Xi(R, \theta, \phi). \tag{2}$$

where ψ_{el} describes the electronic motion and $\Xi(R, \theta, \phi)$ describes the nuclear motion, including both radial and rotational degrees of freedom.

$$\hat{H}_{\mathrm{el}}\psi_{\mathrm{el}} = E_{\mathrm{el}}(R)\psi_{\mathrm{el}}, \tag{3}$$

yielding an electronic energy that depends only on the magnitude of the internuclear separation $R = |\mathbf{R}_A - \mathbf{R}_B|$. This energy defines the potential energy curve (PEC),

$$V(R) = E_{\mathrm{el}}(R),$$

on which the nuclear motion is subsequently solved.

2.1.2 Potential energy curves and rovibrational structure

The potential energy curve describes how the electronic energy of a diatomic molecule varies with the internuclear separation R, forming the potential that governs the vibrational and rotational motion. A schematic example is shown in Figure 1. For bound states, the curve has a minimum at the equilibrium bond length R_e, corresponding to the most stable configuration, and approaches the dissociation energy D_e at large R. Nuclear motion on this potential produces discrete vibrational levels labelled by v, whose spacing decreases with increasing energy due to the anharmonic shape of the potential.

Each vibrational level is further split into rotational levels labelled by J, as illustrated in Figure 2. For a diatomic molecule the angular dependence of nuclear motion can be separated, introducing the rotational quantum number J and reducing the problem to a radial equation whose solutions $\chi_{vJ}(R)$ define the rovibrational energy levels. The rotational spacings are typically much smaller than the vibrational spacings, producing a hierarchical level structure.

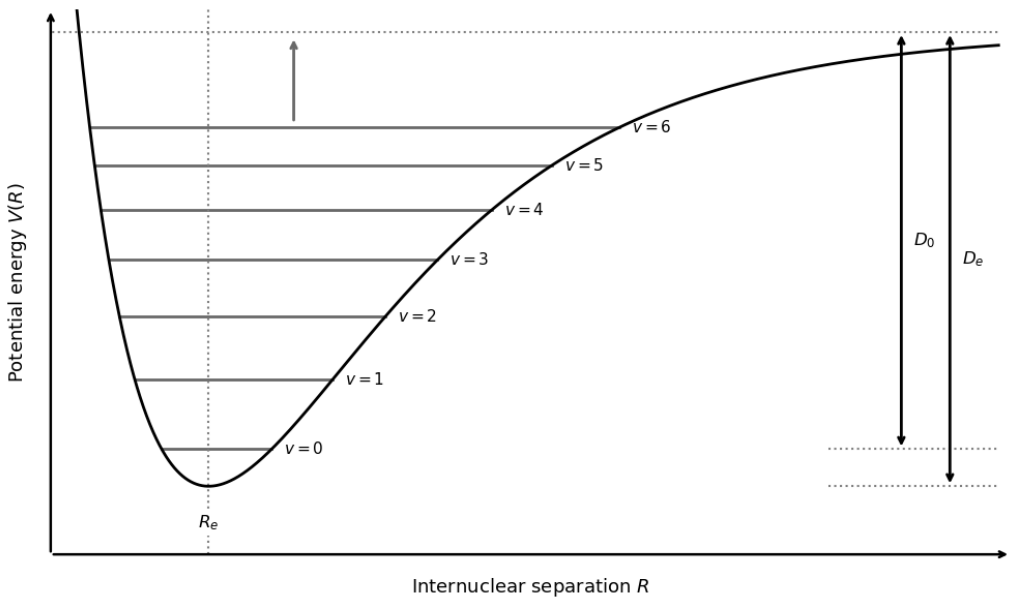

Figure 1: Schematic potential energy curve for a diatomic molecule showing vibrational energy levels $v = 0$–6. The equilibrium bond length R_e, dissociation energy D_e, and zero-point dissociation energy D_0 are indicated. The spacing between vibrational levels decreases with increasing v due to anharmonicity. Only a subset of bound vibrational states is shown; additional levels approach the dissociation limit.

The nuclear motion on the PEC is governed by

$$\left[-\frac{\hbar^2}{2\mu}\frac{d^2}{dR^2} + \frac{J(J+1)\hbar^2}{2\mu R^2} + V(R)\right]\chi_{vJ}(R) = E_{vJ}\chi_{vJ}(R), \tag{4}$$

where μ is the reduced mass of the molecule [33]. The first term represents vibrational kinetic energy, the second term represents rotational kinetic energy, and $V(R)$ is the PEC introduced above.

In a simple approximation, the rovibrational energy levels are

$$E_{vJ} \approx \omega_e\left(v+\tfrac{1}{2}\right) - \omega_e x_e\left(v+\tfrac{1}{2}\right)^2 + B_v J(J+1),$$

where ω_e is the harmonic vibrational frequency, $\omega_e x_e$ the leading anharmonic correction, and B_v the rotational constant for level v. Higher-order terms, including centrifugal distortion, are omitted.

Radiative transitions between these rovibrational levels produce spectral lines, as illustrated in Figure 2. Pure rotational transitions ($\Delta v = 0$) typically occur in the far-infrared, while rovibrational transitions ($\Delta v \geq 1$) lie in the infrared. For most

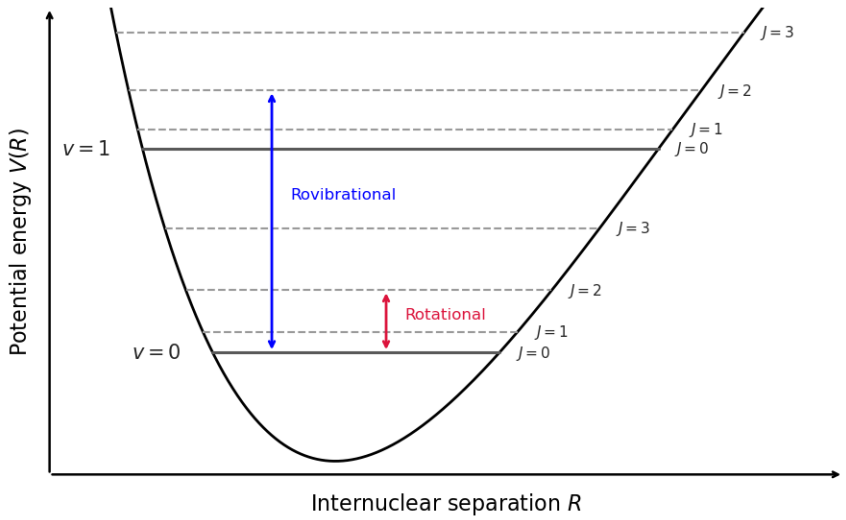

Figure 2: Expanded view of the lower region of the potential energy curve in Figure 1 illustrating rotational substructure within the $v = 0$ and $v = 1$ vibrational states. Rotational levels are labelled by the quantum number J. Example rotational ($\Delta v = 0$) and rovibrational ($\Delta v = 1$) transitions are indicated.

molecules these transitions arise from electric dipole radiation and obey the selection rules $\Delta v = \pm 1$ and $\Delta J = \pm 1$. However, molecular nitrogen (N_2) is homonuclear, possessing two identical nuclei and therefore no permanent dipole moment. Dipole transitions within the ground electronic state are therefore forbidden, and radiative transitions occur instead via the much weaker electric quadrupole interaction ($\Delta J = 0, \pm 2$), typically 10^6 times weaker than dipole transitions [6]. A further consequence is that transitions occur only between even-J or odd-J levels, producing two independent rotational populations (para- and ortho-N_2).

In practical calculations the potential energy curve must be specified analytically or numerically. The PEC adopted for the present work is described below.

2.1.3 Adopted potential energy curve

The PEC used in this work is the recommended $MLR_4(6, 8)$ potential for the ground electronic state of N_2 reported by Le Roy *et al.* [12]. It was obtained by fitting an analytic function directly to an extensive spectroscopic dataset consisting primarily of Raman measurements, together with electric quadrupole rovibrational transitions and term values derived from electronic spectra.

The potential follows the Morse–Long-Range (MLR) functional form, which combines Morse-like flexibility near equilibrium with the correct asymptotic behaviour

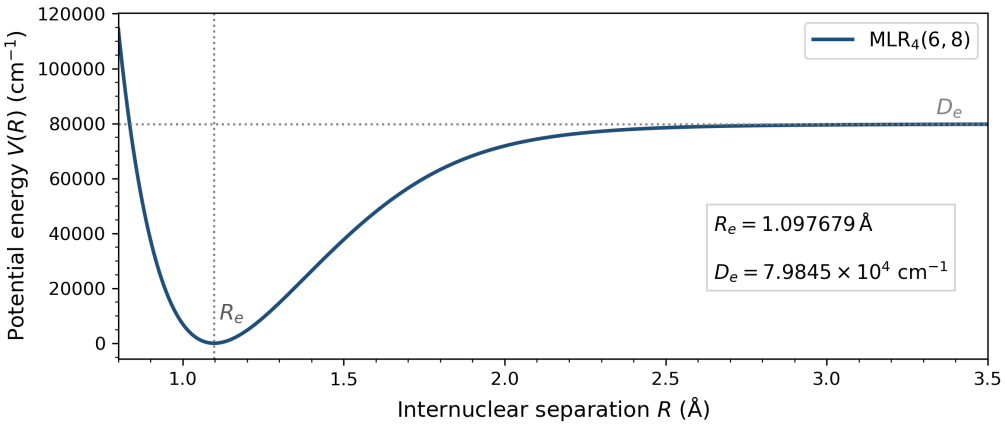

Figure 3: Potential energy curve for $^{14}N_2$ obtained from the fitted MLR$_4(6,8)$ function. The curve is shown over the range 0.8-3.5 Å; a wider grid was used in the analysis.

at large internuclear separation. For N_2 it is written

$$V_{\rm MLR}(R) = D_e \left[1 - \left(\frac{R_e}{R} \right)^n \left(\frac{1 + R_{m,n}/R^{m-n}}{1 + R_{m,n}/R_e^{m-n}} \right) e^{-\phi(R)\, y_p(R)} \right]^2 \tag{5}$$

where n and m specify the inverse-power terms governing the long-range behaviour of the potential (here, $n = 6$, $m = 8$). The exponent function $\phi(R)$ is expressed as a polynomial expansion in the radial variable $y_p(R)$, defined by the Surkus relation

$$y_p(R) = \frac{R^p - R_e^p}{R^p + R_e^p}. \tag{6}$$

with $p = 4$ for the MLR$_4(6,8)$ potential. The resulting potential is shown in Figure 3 with key parameter values; the full fitted parameters and spectroscopic dataset are given in Ref [12].

Wavenumbers $\tilde{\nu}$ (in cm^{-1}) are used throughout this work, as is standard in spectroscopy, and are related to frequency by $\tilde{\nu} = \nu/c$.

Prior to the study by Le Roy *et al.* [12], there was no widely accepted, accurate analytic PEC for the ground state of N_2. Earlier representations, such as Rydberg–Klein–Rees (RKR) potentials [21], reconstruct the curve pointwise from spectroscopic data and therefore extrapolate poorly beyond the fitted region. High-quality *ab initio* calculations have also been reported, such as the r12-MR-ACPF potential of Gdanitz [7], but they do not achieve full spectroscopic accuracy and show increasing discrepancies for higher vibrational levels.

By contrast, the direct-potential-fit (DPF) approach of Le Roy *et al.* produces a potential that reproduces the available spectroscopic dataset used in the fit with high accuracy and is expected to provide reliable predictions for rotational levels with very high J within the vibrational data range ($v < 19$) [12]. The authors further report that it should give "quite good predictions" for even higher vibrational levels beyond the fitted region. This level of accuracy makes the potential well suited for calculating the rovibrational energies required to generate a comprehensive molecular line list for high-temperature applications.

2.1.4 Born–Oppenheimer breakdown correction for isotopologues

Line lists are required not only for $^{14}N_2$, but also for the isotopologues $^{14}N^{15}N$ and $^{15}N_2$. Under the Born–Oppenheimer (BO) approximation the electronic PEC is independent of nuclear mass, so isotopic substitution affects the rovibrational energies only through the reduced mass in Equation 4. In practice, a small isotope-dependent correction to the potential is required, known as the adiabatic Born–Oppenheimer breakdown (BOB) correction [11], which modifies the effective potential for an isotopologue ξ as

$$V^{(\xi)}(R) = V_{\mathrm{MLR}}(R) + \Delta V_{\mathrm{ad}}^{(\xi)}(R). \tag{7}$$

The correction term is given by

$$\Delta V_{\mathrm{ad}}^{(\xi)}(R) = \left(\frac{M_{N_a}^{(\xi)} - M_{N_a}^{(1)}}{M_{N_a}^{(\xi)}} + \frac{M_{N_b}^{(\xi)} - M_{N_b}^{(1)}}{M_{N_b}^{(\xi)}} \right) S_{\mathrm{ad}}(R), \tag{8}$$

where $M_{N_{a,b}}$ are the nuclear masses and superscript (1) denotes the reference isotopologue ($^{14}N_2$). The function $S_{\mathrm{ad}}(R)$ is the adiabatic BOB radial function which, following Ref. [12], is expressed in terms of the Surkus variable introduced above,

$$S_{\mathrm{ad}}(R) = u_1^{\mathrm{N}}\, y_6(R)\, \left(1 - y_6(R)\right), \tag{9}$$

with fitted parameter $u_1^{\mathrm{N}} = -14.1\ \mathrm{cm}^{-1}$.

The MLR potential (Equation 5) was fitted to spectroscopic data for the reference isotopologue $^{14}N_2$ and therefore already represents its effective adiabatic potential. For other isotopologues, only the isotope-dependent correction in Equation 8 must be applied. The practical implementation of these corrections in the nuclear-motion calculations is described in Section 3.1.1.

Nonadiabatic centrifugal BOB corrections, which modify the rotational kinetic-energy operator rather than the PEC, were also tested but had no discernible effect on the fit and were therefore neglected [12].

2.1.5 Non-quadrupole transitions

In addition to electric quadrupole transitions, other radiative mechanisms were considered but ultimately not included in these line lists.

Magnetic dipole transitions: These arise from the molecular magnetic dipole moment rather than an electric multipole moment. The magnetic dipole operator is proportional to $\mathbf{L} + 2\mathbf{S}$, where $\mathbf{L}$ and $\mathbf{S}$ are the electronic orbital and spin angular momenta. For the ground electronic state of nitrogen, $X\,^1\Sigma$, both vanish ($L = 0$, $S = 0$), so the magnetic dipole moment is zero. This remains true for all isotopologues, and magnetic dipole transitions are therefore not expected to contribute to the presented line lists.

Non-BO electric dipole transitions: Within the Born–Oppenheimer (BO) approximation, rovibrational electric dipole (E1) transitions are forbidden for N_2, including the heteronuclear isotopologue $^{14}N^{15}N$. However, the unequal nuclear masses in $^{14}N^{15}N$ slightly break this symmetry, producing a very small effective dipole moment and allowing weak E1 transitions with $\Delta J = \pm 1$. This mechanism is well established in HD, where purely non-BO effects give observable rovibrational transitions [19]. For $^{14}N^{15}N$, the only available estimate is an early calculation by Bates [1], giving a fundamental-band oscillator strength $f_{0\to1} \sim 2 \times 10^{-14}$. Converting this to a band-integrated absorption strength and comparing with summed HITRAN quadrupole intensities (with isotopic abundance scaling removed; see Section 2.3.1) gives a rough ratio $S_{\mathrm{E1}}/S_{\mathrm{E2}} \sim 3$, suggesting that non-BO E1 absorption may be comparable to or stronger than electric quadrupole (E2) absorption for this isotopologue. These transitions are not included in the HITRAN line list discussed in Section 2.3.1, and there is currently no clear observational evidence for them in laboratory or atmospheric spectra, including long-pathlength studies described in Section 2.3.2. Given the approximate nature of the Bates estimate and the absence of a modern dipole moment calculation, such transitions are not included in the present work but may merit future investigation.

Collision-induced absorption (CIA): This occurs when collisions create a temporary dipole moment, allowing absorption by molecules such as N_2 that lack a permanent dipole. Unlike the discrete quadrupole spectrum, CIA forms a broad continuum that can overlap similar spectral regions. CIA scales with pressure squared, whereas quadrupole lines scale linearly with pressure, so CIA can become relatively more important in dense atmospheres. [23]. Figure 4 shows representative HITRAN

CIA coefficients [9] for N_2 with several collision partners, using curves closest to 296 K. Higher-temperature data are limited and available datasets are incomplete, so the figure provides qualitative guidance only; comparison with quadrupole absorption is discussed further in Section 2.3.1.

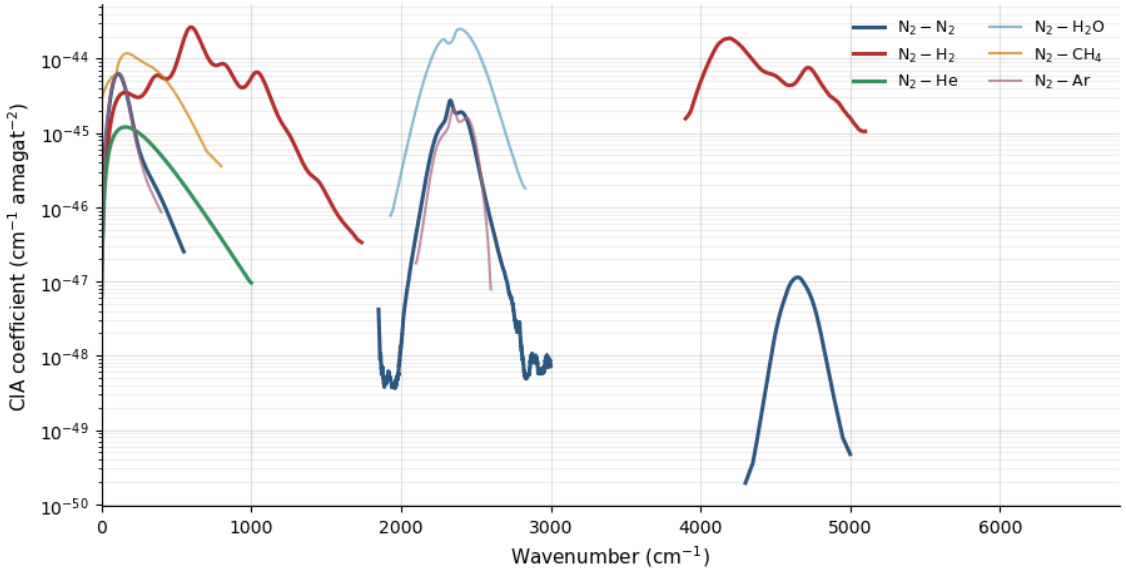

Figure 4: CIA coefficients for N_2 with selected collision partners from HITRAN [9]. The coefficient (cm^{-1} $amagat^{-2}$) gives absorption per molecular pair, where 1 amagat is the ideal-gas number density at standard conditions. Bold curves indicate the most relevant collision partners. Curves correspond to temperatures closest to 296 K, subject to data availability, and have been smoothed for clarity.

2.2 Line intensities

While the PEC determines the rovibrational energy levels and therefore the positions of spectral lines, their intensities depend on the transition probabilities between those levels, governed by the molecular quadrupole moment function.

2.2.1 Transition probabilities

The strength of a spectral line is determined by the probability of radiative transitions between molecular energy levels, commonly expressed through the Einstein A coefficient which gives the spontaneous emission rate from an upper state j to a lower state i. For electric quadrupole transitions this is given by

$$A_{ji} = \frac{64\pi^4\nu^5}{15hc^5(2J_j+1)} \left|\langle\psi_i|Q(R)|\psi_j\rangle\right|^2, \tag{10}$$

where ν is the transition frequency, J_j is the rotational quantum number of the upper state, and $Q(R)$ is the electric quadrupole moment function (QMF). The transition probability therefore depends on the matrix element of $Q(R)$ between the rovibrational wavefunctions ψ_i and ψ_j, obtained from Equation 4. The form of the QMF therefore plays a central role in determining line intensities and constitutes the second key input required to construct the line list, alongside the PEC.

2.2.2 Quadrupole moment function

The electric quadrupole moment function $Q(R)$ describes how the electric quadrupole moment of the molecule—a second-order moment of the charge distribution that measures its deviation from spherical symmetry—varies with internuclear separation R. The function used in this work is taken from the *ab initio* calculations of Li and Le Roy (2007) [13] for the N_2 ground-state. Unlike the potential energy curve, the QMF cannot generally be refined directly from experimental spectra and is therefore obtained from first-principles electronic structure calculations.

The resulting QMF is shown in Figure 5. It varies smoothly with internuclear distance R and approaches zero at large separation, as expected for two neutral atoms, and was calculated on a dense radial grid spanning 0.8–12.1 bohr ($\approx$ 0.42–6.40 Å). Comparison with the PEC in Figure 3 shows that this range fully covers the region relevant for bound rovibrational states, with increased resolution near the equilibrium separation. In this work the tabulated values are used directly rather than fitting an analytic representation; Li–Le Roy show that spline interpolation (as used by the calculation software) converges quadrupole matrix elements to better than 10^{-6} a.u., so numerical effects from interpolation or limited extrapolation beyond the grid are expected to be negligible.

The recommended `MR-AQCC/aug-cc-pCV5Z` model yields an equilibrium quadrupole moment $Q_e \equiv Q(R_e) = -1.1273$ a.u., with estimated uncertainties of approximately 0.3% in Q_e and 0.8% in its first radial derivative [13]. While the paper does not explicitly quantify the expected intensity accuracy, Li and Le Roy show that earlier semiempirical calculations [8] underestimate quadrupole intensities by about 5%. Thus, the adopted QMF provides a reliable basis for evaluating the quadrupole matrix elements $\langle v'J'|Q(R)|vJ\rangle$.

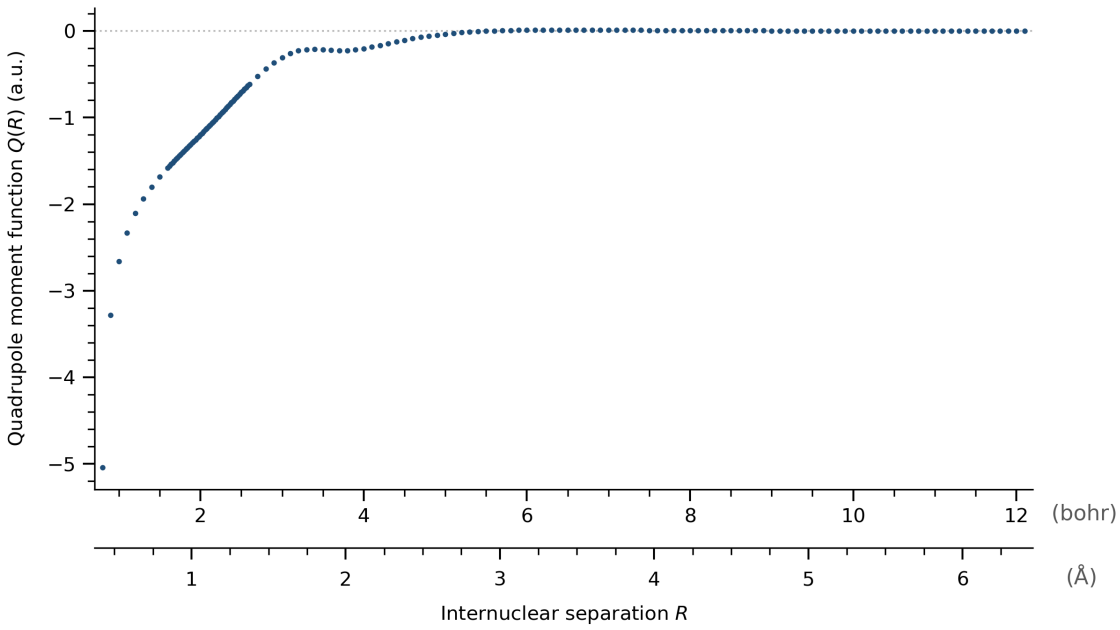

Figure 5: Analytical Li–Le Roy quadrupole moment function $Q(R)$ (in atomic units, a.u. $= ea_0^2$) as a function of internuclear separation R. The function is sampled at 0.1 bohr, with a finer spacing of 0.02 bohr between 1.6 and 2.6 bohr.

2.2.3 From Einstein A coefficients to line intensities

The Einstein A coefficient (Equation 10) describes the probability of spontaneous emission for a transition, but does not by itself determine the observable line intensity. This depends on the energy level populations, which are governed by the thermodynamic conditions of the gas. Under local thermodynamic equilibrium, the population of a state i is given by the Boltzmann distribution,

$$N_i \propto g_i e^{-E_i/kT}, \tag{11}$$

where E_i is the state energy, T the temperature, and g_i the state degeneracy, including rotational and nuclear spin statistical weights. The populations are normalised by the partition function,

$$Q(T) = \sum_i g_i e^{-E_i/kT}, \tag{12}$$

which describes how molecules are distributed over all accessible states.

Combining the transition probability with the Boltzmann population gives the temperature-

dependent line intensity,

$$S_{ij}(T) = I_a \frac{A_{ji}}{8\pi c \tilde{\nu}_{ij}^2} \frac{g_j e^{-E_i/kT}}{Q(T)} \left(1 - e^{-hc\tilde{\nu}_{ij}/kT}\right), \tag{13}$$

where $\tilde{\nu}_{ij}$ is the transition wavenumber and i and j denote the lower and upper states, respectively. This definition follows the HITRAN convention of absorption line intensity [9], in which the lower-state population determines the line strength, with a correction for stimulated emission, and the terrestrial natural abundance I_a is factored in.

Equation 13 links the molecular model described above to observable spectra. The PEC determines the energy levels and transition frequencies, while the QMF determines the transition matrix elements and hence the Einstein coefficients A_{ji}. Together with the partition function, these define the line intensities, so that a line list consists of transition frequencies and Einstein A coefficients from which intensities at any temperature can be obtained. The following section presents the existing line lists for the N_2 ground-state and highlights the characteristic band structure of the quadrupole spectrum.

2.3 Existing line lists and observational context

2.3.1 Existing HITRAN line lists

The HITRAN database [9] is the standard reference source of molecular spectroscopic data, providing transition frequencies, Einstein A coefficients, line intensities, and associated quantities (like self-broadening parameters). It is regularly updated as new experimental and theoretical results become available, and is primarily designed for high-resolution modelling of radiative transfer in planetary atmospheres, particularly that of Earth. As such, the database is tabulated at a reference temperature of 296K (approximately room temperature) and is highly reliable for terrestrial conditions, but may lack completeness at the higher temperatures encountered in hot exoplanet atmospheres and cool stars. At high temperatures, a far larger number of rovibrational states contribute to the spectrum, requiring more complete line lists, which is the central objective of the ExoMol project [31].

For molecular nitrogen, the HITRAN quadrupole line list is based on the same underlying molecular model used in this work: the MLR_4 PEC of Le Roy *et al.* and the QMF of Li and Le Roy. Prior to these calculations, N_2 quadrupole intensities were estimated using semi-empirical expressions fitted to limited experimental data [8], rather than from a consistent PEC–QMF model applicable across the full spectrum. Quadrupole transitions for $^{14}N_2$ and $^{14}N^{15}N$ were first included in HITRAN2008, with a subsequent correction to the line intensity calculations in HITRAN2012; no

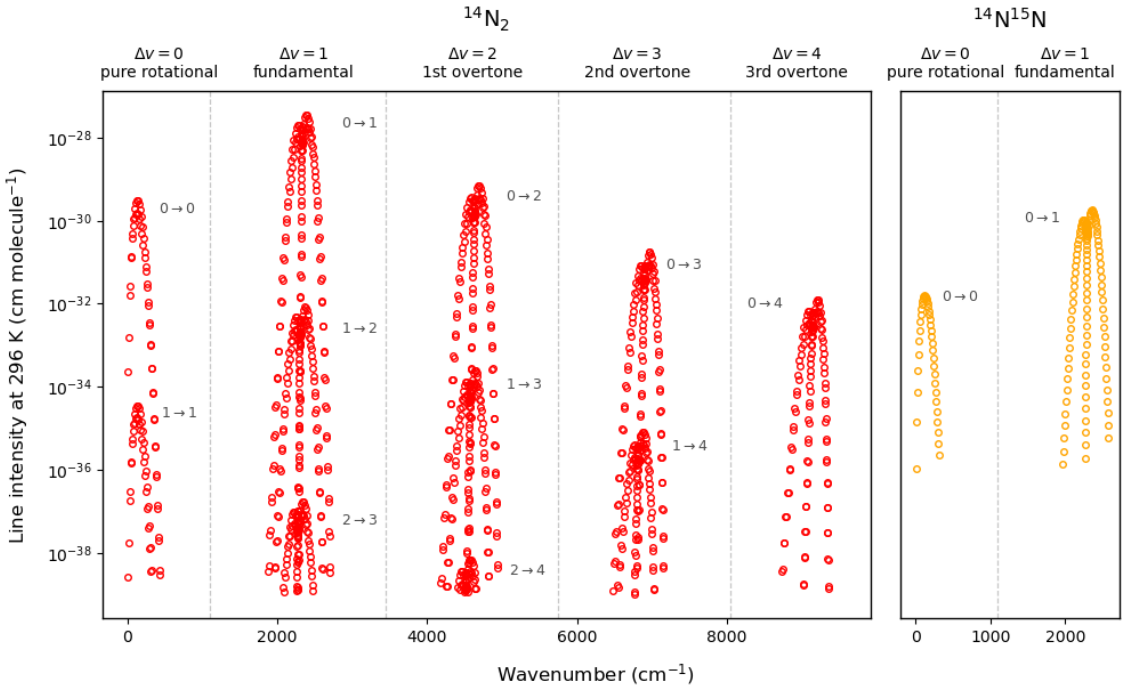

Figure 6: HITRAN2024 quadrupole line intensities for $^{14}N_2$ (left) and $^{14}N^{15}N$ (right) at 296 K, spanning the far-IR to near-IR region, shown as stick spectra coloured by isotopologue. Transitions are grouped into vibrational bands labelled by $v'' \to v'$, with vertical lines separating regions of different Δv. Intensities are plotted on a logarithmic scale in units of cm molecule^{-1} corresponding to the integrated absorption cross-section per molecule.

further updates have been made since. The line lists for $^{14}N_2$ (left) and $^{14}N^{15}N$ (right) are visualised in Figure 6. Notably, quadrupole transitions for $^{15}N_2$ are entirely absent from the database.

Focusing first on $^{14}N_2$ in Figure 6, the transitions are clearly grouped into bands according to the change in vibrational quantum number v. The pure rotational band ($\Delta v = 0$) appears at low wavenumber (< 200 cm^{-1}), while the fundamental band ($v = 0 \to 1$) forms the strongest feature near 2350 cm^{-1}, and is therefore the most likely to be detected. In Section 2.1.2, the selection rule was introduced as $\Delta v = 1$, which follows from the harmonic oscillator approximation. Real molecular potentials exhibit anharmonicity, allowing weak but non-zero transition probabilities for $\Delta v = 2, 3, 4$. These overtone bands are visible in the plot at progressively higher wavenumbers with decreasing intensity. Within each vibrational band, many individual lines arise from transitions between different rotational levels J, producing the structured appearance visible in Figure 6. The pattern differs slightly from

dipole spectra, reflecting the quadrupole selection rule $\Delta J = \pm 2$. In addition to transitions from the ground vibrational state, weaker hot bands (e.g. $v = 1 \rightarrow 2$, $2 \rightarrow 3$) are present, arising from thermally populated excited vibrational levels. The intensities shown correspond to 296 K; at higher temperatures, hot bands and higher-J transitions become more populated, producing a denser spectrum.

The existing HITRAN line list for $^{14}N_2$ is relatively extensive at 296 K, including the fundamental band, several overtone bands, and associated hot bands, with intensities down to $\sim 10^{-39}$ cm molecule^{-1}. For higher-temperature applications, additional hot bands and higher-J states may contribute significantly, although for this isotopologue the primary aim is to reproduce the existing HITRAN calculations within an independent computational framework. The situation is considerably more limited for $^{14}N^{15}N$, for which only the fundamental band and pure rotational transitions are included, with no overtone or hot-band coverage and a reduced range of J values. Quadrupole transitions for $^{15}N_2$ are entirely absent. Extending the line list to higher excitation and to all stable isotopologues therefore enables more reliable modelling of high-temperature planetary and stellar atmospheres, including environments with isotopic abundances differing from those on Earth. The aim of this work is to produce line lists that are as complete as possible within the limits of the available PEC and QMF and numerical noise.

Figure 4 shows that CIA produces a broad continuum overlapping similar spectral regions to the quadrupole bands. As the quantities and available temperature ranges differ, only a qualitative comparison is presented here. In hot, high-pressure atmospheres, such as hot Jupiters, N_2–N_2 (self-CIA), N_2–H_2, and N_2–He are expected to dominate; hydrogen may obscure much of the pure rotational band and part of the first overtone, while self-CIA mainly affects the rotational region and more weakly the fundamental. Methane and water can contribute additional continuum opacity in overlapping regions, potentially complicating identification of the weakest quadrupole features.

2.3.2 Existing observations

Quadrupole transitions of N_2 have been experimentally measured, although most observations have been performed under controlled laboratory conditions due to their intrinsic weakness. The MLR_4 PEC was refined with high-precision Fourier Transform Spectroscopy measurements of a small number of $v = 0 \rightarrow 1$ fundamental-band transitions: 18 lines measured in the laboratory by Reuter *et al.* (1986) [22] and 19 lines retrieved from aircraft-based solar occultation spectra by Rinsland *et al.* (1991) [24]. More recently, advances in cavity ring-down spectroscopy have enabled detection of extremely weak overtone transitions, including a first-overtone line ($v = 0 \rightarrow 2$) measured by Čermák *et al.* (2017) [35] and two second-overtone

lines ($v = 0 \rightarrow 3$) reported by Kassi *et al.* (2013) [10], with intensities on the order of 10^{-31} cm molecule^{-1}, among the weakest molecular absorption features yet measured in the laboratory. Although not included in the original fit, agreement of these overtone measurements with predicted line positions and intensities provides independent support for the PEC–QMF model at larger internuclear separations.

Evidence that N_2 quadrupole absorption contributes measurably to atmospheric spectra has also been obtained from space-based observations. Analysis of solar occultation spectra recorded by the ACE (Atmospheric Chemistry Experiment) satellite showed that inclusion of N_2 line-by-line absorption reduces residuals in the 1900–2800 cm^{-1} region [26], corresponding to the $v = 0 \rightarrow 1$ fundamental band. Individual quadrupole transitions are not resolved, but the band contribution is required to accurately reproduce the observed spectrum.

Taken together, laboratory measurements now probe extremely weak overtone transitions, supporting the PEC–QMF description in a regime not included in the original quadrupole dataset, while ACE observations show that N_2 quadrupole absorption produces a measurable band-level effect in atmospheric spectra. Existing line lists are generally sufficient given current observational capabilities, but improving instrumental sensitivity motivates extending calculations to higher vibrational and rotational excitation, particularly where isotopologue coverage remains incomplete.

2.3.3 Key information for molecular nitrogen

For reference, Table 1 summarises basic molecular properties of the stable isotopologues of N_2 relevant to the line list calculations that will follow.

Table 1: Properties of the stable isotopologues of N_2.

	$^{14}N_2$	$^{14}N^{15}N$	$^{15}N_2$
Nuclear spin I	1, 1	1, $\frac{1}{2}$	$\frac{1}{2}$, $\frac{1}{2}$
Term symbol	$X\,^1\Sigma_g^+$	$X\,^1\Sigma^+$	$X\,^1\Sigma_g^+$
$g_{\rm ns}$ (even J)	6	6	1
$g_{\rm ns}$ (odd J)	3	6	3
Molecular mass (amu)	28.006	29.003	30.000
Reduced mass μ (amu)	7.000	7.241	7.500
Natural abundance I_a	9.92687×10^{-1}	7.29916×10^{-3}	1.34176×10^{-5}

Natural abundances I_a follow the recommended terrestrial ratio $^{14}N/^{15}N = 272$ [4], consistent with HITRAN values. Molecular masses are from NIST [5]. Here $g_{\rm ns}$ is the nuclear spin statistical weight, arising from symmetry constraints on the total wavefunction. These determine the relative populations of even- and odd-J levels,

affecting the line intensities. The total degeneracy in Eq. 12 is $g_i = g_{\rm ns}(i)(2J_i + 1)$, which incorporates the rotational degeneracy. The heteronuclear isotopologue $^{14}N^{15}N$ lacks inversion symmetry and therefore has no gerade label (^{g}).

3 Methodology

To obtain comprehensive rovibrational line lists for all three isotopologues of N_2, the nuclear-motion Schrödinger equation was solved using the program Duo, a general rovibronic solver for diatomic molecules developed within the ExoMol project [33]. `Duo` solves the one-dimensional nuclear-motion problem variationally and computes transition frequencies and Einstein A coefficients from user-specified potential energy and property curves.

Since the same PEC and QMF are adopted as in HITRAN, close agreement is expected with both the HITRAN line positions and the energy levels reported by Le Roy *et al.* [12]. Comparison with these reference values provides a useful check of the implementation, with any small differences expected to arise from numerical aspects of the calculation, such as basis-set truncation and grid discretisation.

The Duo input files used in this work are provided via the repository linked to in the Appendix to ensure reproducibility.

3.1 Energy levels and transition frequencies

Rovibrational energy levels were computed in Duo using the adopted PEC, allowing direct comparison with published values reported by Le Roy *et al.* [12]. Calculations were first performed for purely vibrational levels ($J = 0$) before extending to a wide range of vibrational and rotational quantum numbers (v, J). The resulting transitions can subsequently be refined by applying filters to exclude those significantly affected by numerical noise.

3.1.1 Defining the PEC in Duo

The first step is to define the PEC in Duo, which supports several pre-defined analytical functional forms. Its Morse Long-Range (MLR) representation is equivalent to the MLR_4 form given in Equation 5.

This allows the $^{14}N_2$ potential to be implemented analytically. However, the other isotopologues require inclusion of the adiabatic BOB correction described in Section 2.1.4, which is not currently supported analytically within Duo. The BOB-corrected PEC was therefore evaluated on a numerical grid in Python and imported into Duo on the same radial grid used in the Duo calculations, avoiding interpolation.

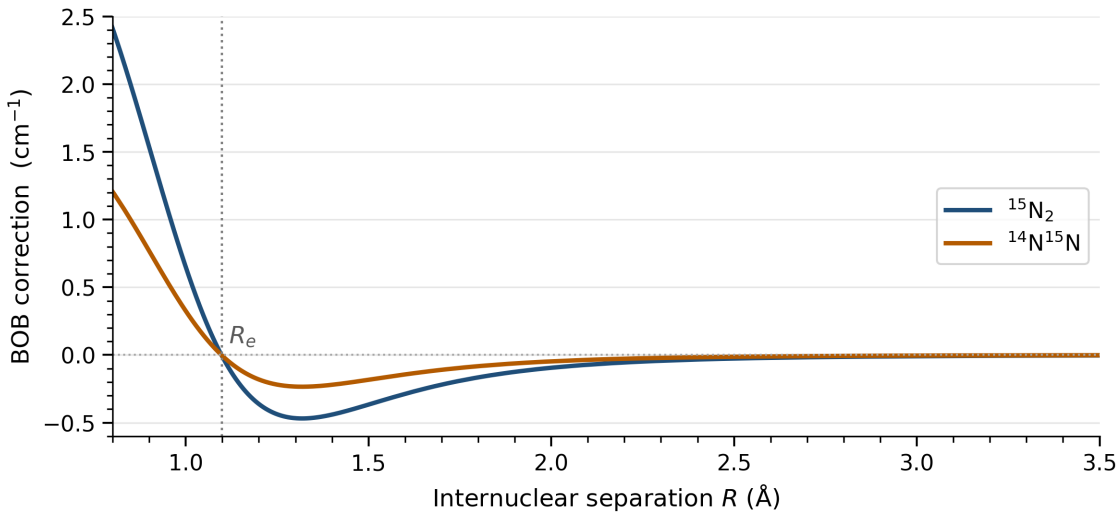

Figure 7: Adiabatic Born–Oppenheimer breakdown (BOB) correction to the $^{14}N_2$ potential energy curve for $^{14}N^{15}N$ and $^{15}N_2$, shown as the difference obtained by subtracting the uncorrected Born–Oppenheimer potential from the BOB-corrected potential as a function of internuclear separation R.

The effect of the BOB correction is shown in Figure 7. The correction is zero at R_e, negative at larger separations and positive at shorter separations. Since higher vibrational states sample larger average internuclear distances (Figures 3 and 6), the correction is therefore expected to be negative for the bands considered here. Its magnitude is of order $0.1\,\mathrm{cm}^{-1}$ and is therefore non-negligible at spectroscopic precision.

Calculating the PEC independently in Python introduces a small numerical discrepancy relative to Duo for numerical reasons. The relative error, of order 10^{-9}, is negligible compared to spectroscopic uncertainties and has no significant effect on calculated energy levels. Due to the discrete grid representation, the minimum of the PEC is slightly offset from zero; this is handled within Duo by disabling automatic shifting of the potential, allowing direct comparison with literature values.

3.1.2 Computing the energy levels

Energy levels were computed using a linear radial grid spanning a wide range of internuclear separations ($R = 0.2$–12 Å, 7001 points). Calculations were first performed for purely vibrational states ($J = 0$). The lowest five vibrational levels were computed initially to verify correct implementation of the PEC and Duo input, reproducing published values to all four decimal places [12]. Calculations were then extended to higher vibrational states.

To obtain convergence for levels approaching the dissociation energy D_e (Figure 3), the radial grid range was increased substantially. Owing to the relative simplicity of the present line list, a large linear grid could be employed while maintaining sufficiently fine spacing to avoid loss of numerical accuracy. Le Roy *et al.* restricted calculations to levels below dissociation, corresponding to $v_{\max} = 60, 61$, and 62 for $^{14}N_2$, $^{14}N^{15}N$, and $^{15}N_2$, respectively. Here, levels were computed up to $v = 80$ to ensure a sufficiently complete vibrational basis; those near dissociation can later be excluded if affected by numerical noise. Agreement with published values was maintained across this extended range for all three isotopologues, so a comparison table is not reproduced here.

Once convergence with respect to v was established, calculations were extended to include rotational excitation up to $J = 80$, again chosen to ensure broad coverage of the (v, J) space. While $J = 0$ levels for a given v depend only on the corresponding vibrational basis function, calculations with $J > 0$ require inclusion of higher-v basis states in order to account for centrifugal distortion of the effective potential. This provides further justification for the extended vibrational basis adopted here. Comparison with published values is discussed in the following section.

3.1.3 Computing the transition frequencies

Transition wavenumbers were obtained in Duo as differences between computed upper and lower energy levels, with allowed transitions determined automatically from the quadrupole selection rules. Matching to HITRAN was performed in Python by comparing decoded quantum numbers.

For $^{14}N_2$, all 1071 HITRAN transitions were recovered, agreeing to six decimal places, with residual differences of $10^{-6}\,\mathrm{cm}^{-1}$ attributable to round-off error in the energy levels.

For $^{14}N^{15}N$, all 161 HITRAN transitions were also recovered, but small systematic discrepancies of order $10^{-3}\,\mathrm{cm}^{-1}$ were observed, increasing with J. Differences are smallest for low-J rotational transitions and larger for the fundamental band. Since the computed $J = 0$ vibrational levels reproduce those of Le Roy *et al.* [12] up to dissociation, the discrepancies most likely originate from the HITRAN reference values, potentially reflecting a more limited basis set or omission of the BOB correction in the HITRAN calculation.

3.1.4 Estimating uncertainties in transition frequencies

Uncertainties were assigned to the calculated energy levels using a simple empirical model based on the precision and coverage of the fitted spectroscopic data [12].

The most accurate measurements, including Raman rotational and fundamental-band spectra, have uncertainties of order $10^{-3}\,\mathrm{cm}^{-1}$, while hotter-band data are somewhat less precise. Since the recommended $\mathrm{MLR}_4(6,8)$ potential reproduces the dataset with $dd = 1.44$ (dimensionless RMS fit deviation), a baseline uncertainty of $0.002\,\mathrm{cm}^{-1}$ was adopted.

Uncertainty was increased above $v = 7$, and more rapidly for $v > 19$, which lies beyond the experimental vibrational range included in the fit. The fitted data extend to $J \sim 48$, and the potential is reported to remain reliable at very high J within the vibrational data range; the rotational contribution to the uncertainty was therefore kept small. A term proportional to $J(J+1)$ was included to allow a gradual increase with rotational excitation, following the functional form adopted in the ExoMol SiO line list [34].

The adopted model for the uncertainty in a level energy was

$$\sigma_E(v, J) = 0.002 + a\,v + bJ(J+1), \tag{14}$$

where $b = 10^{-5}\,\mathrm{cm}^{-1}$, and

$$a = \begin{cases} 0, & v \le 7 \\ 0.01, & 7 < v \le 19 \\ 0.02, & v > 19, \end{cases} \qquad [\mathrm{cm}^{-1}].$$

The same model was applied to all isotopologues, which were derived from the same combined-isotopologue fit including BOB corrections [12].

The uncertainty in a transition wavenumber was then obtained by combining upper and lower level uncertainties in quadrature,

$$\sigma_{\tilde{\nu}} = \sqrt{\sigma_E(v', J')^2 + \sigma_E(v'', J'')^2}. \tag{15}$$

3.2 Transition intensities

The quadrupole moment function (QMF) was imported into Duo on the grid shown in Figure 5, from which Einstein A coefficients were computed. An additional factor of $\frac{\sqrt{6}}{3}$ was applied to convert the Cartesian quadrupole moment to the tensorial form used by Duo [27]. Einstein A coefficients were converted to line intensities using ExoCross [32] with natural isotopic abundances for comparison with HITRAN. Input files are provided in the repository linked to in the Appendix.

Rigorous intensity uncertainties are difficult to quantify because they depend primarily on the accuracy of the *ab initio* quadrupole moment function and its variation with molecular geometry [30], for which limited experimental constraints exist. Consequently, individual line uncertainties have not been assigned. The QMF of Li and Le Roy is smooth and well characterised over the range of internuclear distance sampled by low-v states (Figure 5), supporting reliable matrix elements for the strongest transitions. Higher vibrational states probe a wider range of internuclear distance and are therefore more sensitive to the behaviour of the QMF away from equilibrium, which should lead to higher uncertainties.

At this stage, the only filter applied is a very weak intensity threshold, corresponding to approximately 10^{-50} cm molecule^{-1}, over twenty orders of magnitude below the strongest transitions and far below any experimentally detectable intensity. The resulting line lists, shown in Figure 8, therefore include a very broad set of transitions extending towards dissociation, many of which are affected by numerical noise and will be discussed and filtered in the next section.

Comparison with HITRAN shows strong overall agreement in transition intensities. For $^{14}N_2$, HITRAN intensities are systematically higher by approximately 0.2%, while for $^{14}N^{15}N$ agreement is much closer, typically better than 0.01%. These small discrepancies could be due to the numerical implementation or may reflect minor differences in partition functions (as discussed in Section 4.2.1), but overall agreement at the sub-percent level provides confidence in the implementation.

3.3 Validation and filtering of computed line lists

3.3.1 Computed line lists

The unfiltered line lists for all three isotopologues are shown in Figure 8. They are substantially more complete than HITRAN, containing many additional overtone and hot-band transitions extending to higher wavenumbers and lower intensities.

The overall structure is consistent across isotopologues, as expected, with the $^{15}N_2$ line list showing similar features but at lower intensity, reflecting its lower natural abundance and confirming correct implementation.

Although transitions are obtained up to dissociation, not all are reliable. A consistent decrease in intensity is expected for successive overtones, but this behaviour begins to break down above approximately $20,000\,\mathrm{cm}^{-1}$, indicating the onset of dominant numerical noise. Filtering is therefore required to define a physically reliable subset of the line list, as described below.

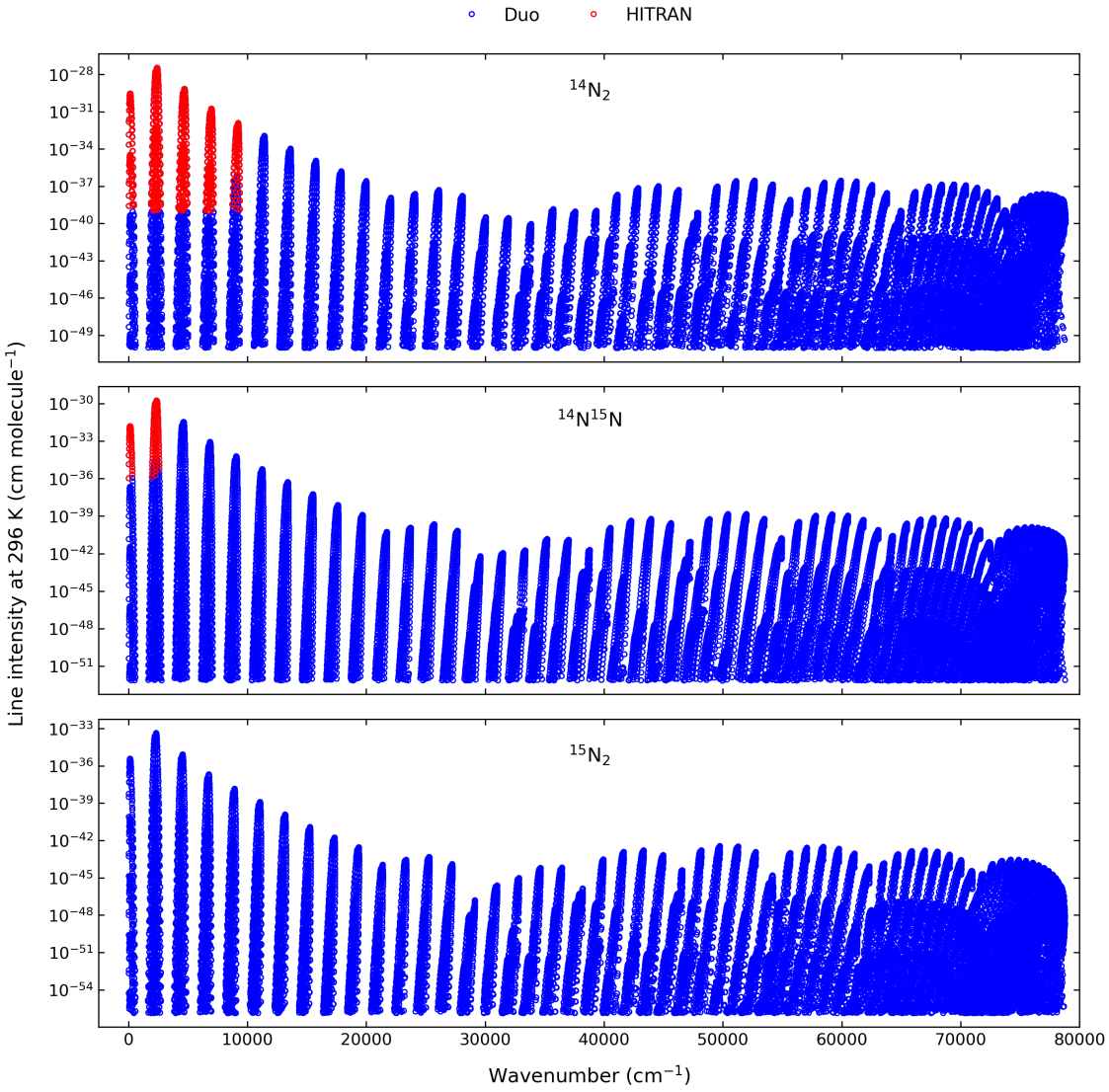

Figure 8: Line intensity plots of the unfiltered Duo quadrupole line lists, computed using rovibrational levels up to $v = 80$ and $J = 80$, at 296 K. The line lists include all transitions written by Duo, subject only to a minimum intensity threshold. HITRAN data, where available, are overlaid in red above the Duo data in blue.

3.3.2 Filtering based on overtone behaviour

The breakdown in intensity behaviour above 20 000 cm^{-1} can be understood from the expected scaling of overtone intensities. High-overtone intensities decrease approximately exponentially with increasing vibrational quantum number difference Δv, so that $\log I \propto -\Delta v$ [14].

This behaviour is observed in Figure 8 up to approximately the $9 \rightarrow 0$ band, beyond which intensities flatten and scatter. Such deviations occur when transition moments become extremely small and numerical contributions become comparable to the physical matrix elements, particularly when the moment function is represented by interpolation on a grid [15], as is the case here. Analytical representations of the QMF can reduce these artefacts and extend the reliable overtone range [15], but were not explored here. Transitions beyond this point were therefore treated as unreliable and excluded by thresholding.

Plotting the vibrational quadrupole matrix elements $|\langle 0|Q|v\rangle|$ (Figure 9) confirms the same behaviour: values decrease smoothly with v up to $v \approx 9$, beyond which they plateau, indicating that numerical noise has begun to dominate. The line list was therefore truncated by band, removing all transitions involving $v \geq 10$, corresponding to a vertical cutoff in Figure 8. This excludes the $10 \rightarrow 0$ band (the ninth overtone) and all higher overtones, together with associated hot bands. An alternative approach would be to filter transitions individually using a threshold on the quadrupole matrix elements; however, this refinement was not explored here.

A global intensity threshold of 10^{-45} cm molecule^{-1} at 296 K was also applied for all isotopologues. This lies far below foreseeable experimental sensitivity, so larger uncertainties in extremely weak transitions have negligible impact on modelling, while preserving completeness and retaining transitions that are weak at the 296 K reference temperature but may become significantly stronger at elevated temperatures.

Summary statistics of the line lists before and after filtering are provided in Table 2.

Table 2: Comparison of ExoMol line lists before and after filtering.

Isotopologue	$^{14}N_2$		$^{14}N^{15}N$		$^{15}N_2$	
	Before	After	Before	After	Before	After
Number of lines	30462	3705	30620	3070	31541	1898
Max wavenumber	78817	20025	78764	19711	78755	19391
Max v	80	9	80	9	80	9
Max J	78	69	79	66	80	59

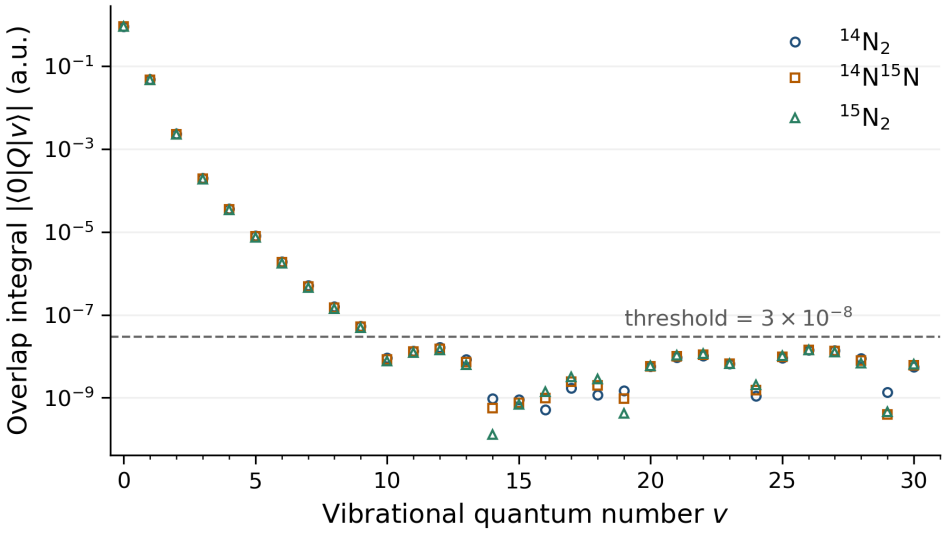

Figure 9: Vibrational matrix elements $|\langle 0|Q|v\rangle|$ of the QMF for the three N_2 isotopologues. The dashed line indicates the threshold of 3×10^{-8} a.u., where decay ceases to be smooth and numerical noise dominates.

4 Results & Discussion

Final rovibrational line lists were produced for all three stable isotopologues of N_2, providing consistent transition frequencies and intensities over a wide range of vibrational and rotational excitation. This section presents an overview of the final line lists, together with temperature-dependent spectra and partition functions demonstrating completeness and suitability for high-temperature atmospheric modelling.

4.1 Final line lists and comparison with HITRAN

Figure 10 compares the calculated ExoMol line lists with HITRAN at 296 K, showing excellent agreement for all isotopologues. As discussed in Sections 3.1.3 & 3.2, all intensities match to sub-percent accuracy, with only minor discrepancies in wavenumbers observed for $^{14}N^{15}N$. The present calculations extend substantially beyond the HITRAN coverage, both in vibrational and rotational excitation.

Table 3 summarises the size of each line list in comparison to HITRAN. For $^{14}N_2$, the number of bands is doubled and the total number of transitions increases threefold, extending to higher J. For $^{14}N^{15}N$, the number of transitions increases by over an order of magnitude, with overtone and hot-band transitions included where none were previously available. A complete line list is also provided for $^{15}N_2$, for which no HITRAN data currently exist.

The final line lists therefore provide substantially more complete coverage at elevated temperatures, where populations of higher-energy rovibrational states become significant, as demonstrated in the following section.

Table 3: Comparison of HITRAN and filtered ExoMol line lists.

Isotopologue	$^{14}N_2$		$^{14}N^{15}N$		$^{15}N_2$	
	HITRAN	ExoMol	HITRAN	ExoMol	HITRAN	ExoMol
Number of lines	1107	3705	161	3070	-	1898
Max wavenumber	9354	20025	2578	19711	-	19391
Max v	4	9	1	9	-	9
Max J	56	69	42	66	-	59

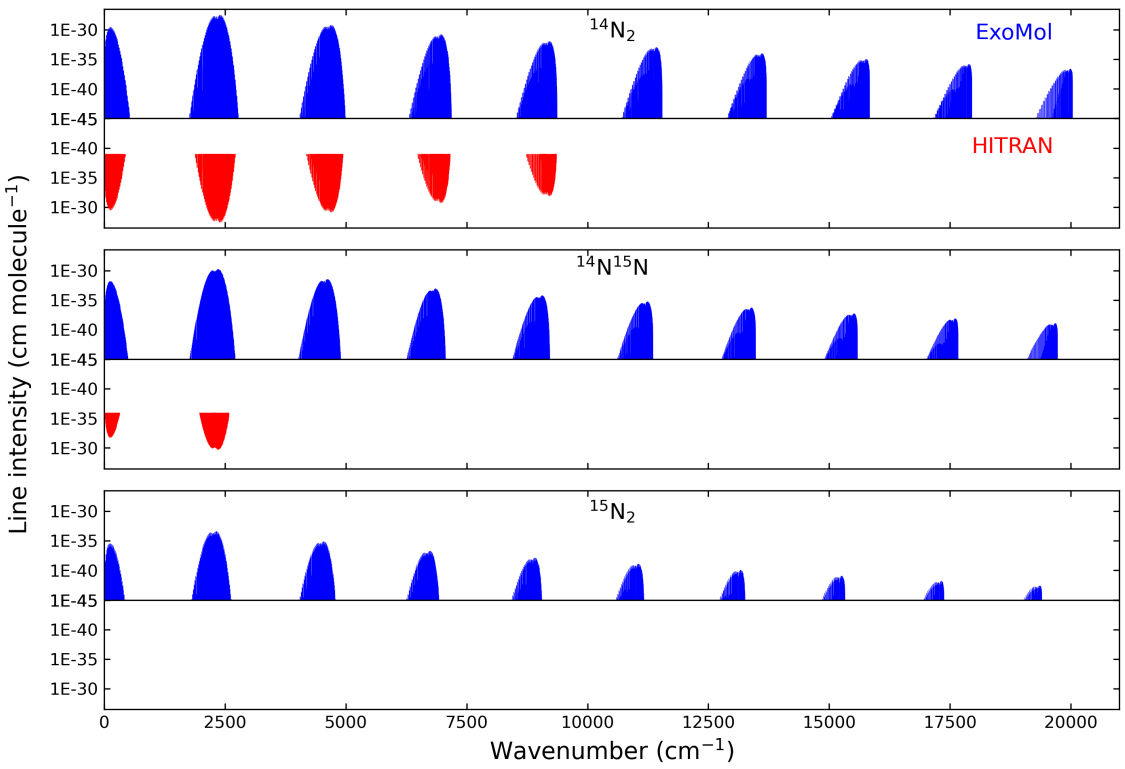

Figure 10: Mirrored stick spectra comparing the final filtered ExoMol line lists (above) with HITRAN (below) at 296 K for the three N_2 isotopologues. The ExoMol spectra correspond to the filtered subset of the unfiltered line lists shown in Figure 8.

4.2 Temperature-dependent spectra

The N_2 quadrupole spectrum varies significantly with temperature as higher rotational states and hot bands become populated. Spectra across a wide temperature range therefore help assess the need for highly excited rovibrational states in the line list, while also allowing isotopic shifts between isotopologues to be examined in high-resolution simulations.

Nitrogen is expected to be present across a wide range of astrophysical environments spanning several orders of magnitude in temperature. At low temperatures ($\sim$50 K), chemical models of protoplanetary disks predict that a major nitrogen reservoir exists in the form of N_2, which remains gaseous interior to the N_2 snowline at tens of AU [20]. At 296 K, N_2 is the dominant constituent of Earth's atmosphere (78% by volume), and this temperature is adopted as the standard reference temperature in HITRAN. At $\sim$1500 K, representative of hot exoplanet atmospheres, thermochemical equilibrium calculations predict a transition in the dominant nitrogen reservoir from NH_3 to N_2 at around 600–800 K [2], consistent with models of hot Jupiters [17]. At $\sim$3000 K, relevant to ultra-hot rocky exoplanets ("lava worlds"), dayside temperatures are observationally inferred to approach 3000 K [18], while thermochemical calculations show that N_2 remains the dominant nitrogen-bearing species up to $\sim$3200 K due to its strong triple bond stability [25]. This temperature regime also encompasses hot brown dwarfs and late M dwarfs, suggesting that nitrogen remains chemically relevant and potentially observable in these objects.

Figure 11 shows Gaussian-broadened spectra of the first three bands at these four representative temperatures. Increasing temperature redistributes intensity across many transitions, producing broader and flatter band structure through population of higher rotational states and hot bands, making band features less distinct and potentially reducing detectability. The relative intensity of the fundamental and overtone bands changes little with temperature, whereas the pure rotational band becomes notably stronger and shifts to higher wavenumber as higher-J states are populated.

Figure 12 shows a higher-resolution view of a narrow region of the fundamental band, with isotopologue contributions shown separately. The increased hot-band contributions at higher temperatures are clearly visible, producing greater line density and a flatter spectral structure. Isotopic shifts between $^{14}N_2$ and $^{14}N^{15}N$ create distinct line positions, introducing additional spectral structure. Retrieval of isotopologue ratios would require extremely high spectral resolution and sensitivity, and is therefore expected to be challenging.

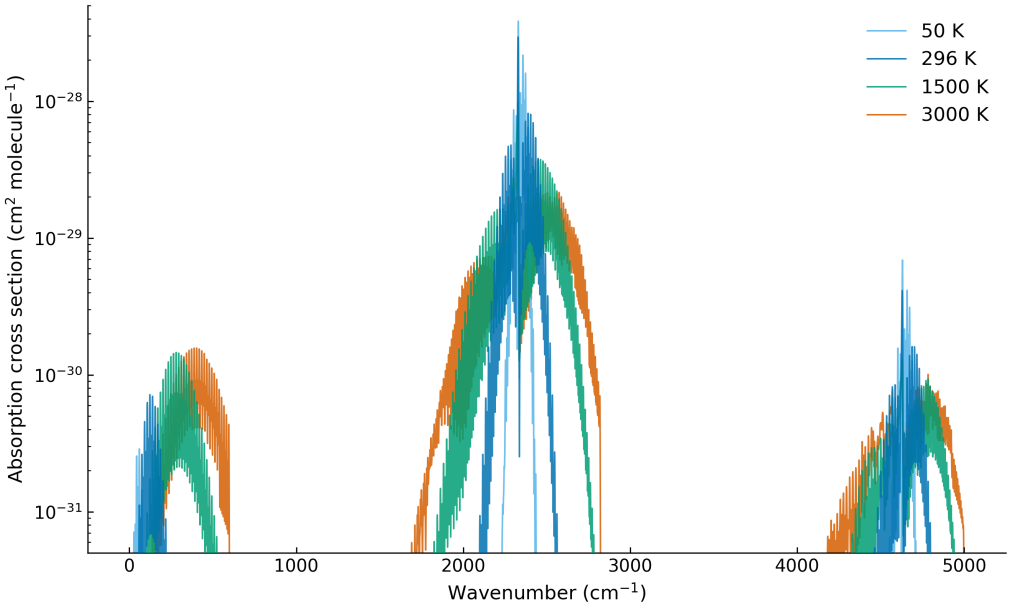

Figure 11: Simulated absorption cross sections on a logarithmic scale for the rotational, fundamental, and first overtone bands. Contributions from all isotopologues are summed, and the spectra are Gaussian-broadened with HWHM = 2 cm^{-1}. Spectra are shown for 50, 296, 1500, and 3000 K to illustrate astrophysically relevant temperature regimes.

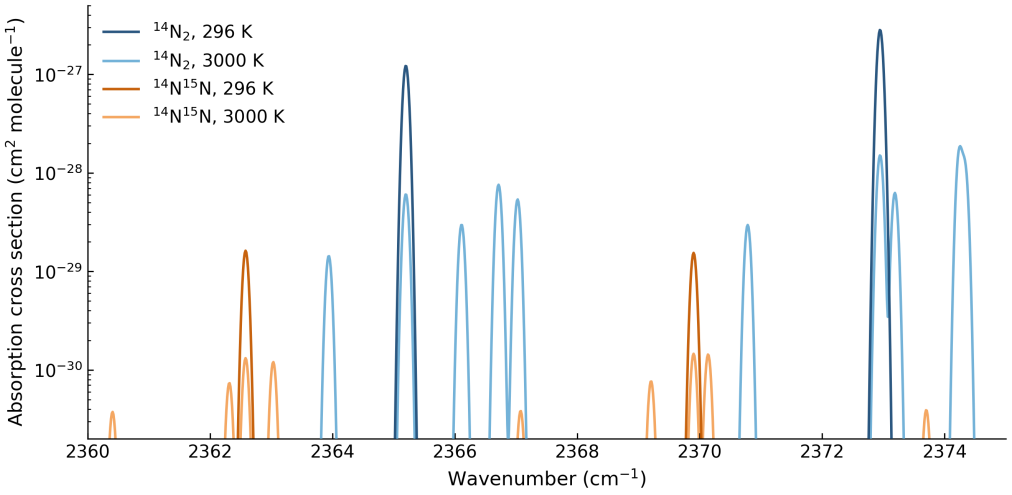

Figure 12: Simulated absorption cross sections for $^{14}N_2$ and $^{14}N^{15}N$ in the 2360–2375 cm^{-1} region (within fundamental band), on a logarithmic scale for 296 and 3000 K. The spectra were Gaussian-broadened with HWHM = 0.5 cm^{-1}, with Doppler broadening neglected.

4.2.1 Partition functions

The partition function provides a useful check on the completeness of the retained rovibrational energy levels, particularly for high-temperature applications where many excited states contribute. Comparison with external partition function datasets therefore helps assess whether the present line list remains sufficiently converged for high-temperature applications.

Figure 13 compares the present results with partition functions from HITRAN and the high-temperature ESA STR-246 compilation [3] over the range 1–9000 K. The HITRAN partition functions are computed from tabulated rovibrational energy levels and are not limited by the transitions included in the HITRAN line lists. The ESA values were scaled by the nuclear spin degeneracy factor $g_{\rm ns} = 9$ for $^{14}N_2$ to match the HITRAN/ExoMol convention. The ESA dataset includes contributions from excited electronic states and is therefore expected to exceed purely ground-state values at sufficiently high temperature.

Excellent agreement is observed between the present results and HITRAN at low temperature, with differences of only 1.5×10^{-4} % at 296 K. This confirms that the small intensity discrepancies discussed in Section 3.2 cannot be attributed to differences in partition function normalisation.

At higher temperatures, the present partition function lies above HITRAN but below the ESA values, indicating improved convergence of the rovibrational energy level basis while remaining physically consistent with the expected electronic contributions included in STR-246. The resulting partition functions are therefore suitable for modelling high-temperature atmospheres and are provided for all three isotopologues, extending beyond the limited isotopologue coverage available in existing databases.

5 Conclusion

This work has produced comprehensive rovibrational line lists for electric quadrupole transitions within the ground electronic state of all three stable isotopologues of N_2: $^{14}N_2$, $^{14}N^{15}N$, and $^{15}N_2$. The calculations employ the recommended Le Roy *et al.* $MLR_4(6,8)$ potential energy curve and Li–Le Roy quadrupole moment function within the Duo and ExoCross framework. While based on the same molecular model as HITRAN, the present work consistently treats all isotopologues and extends coverage to higher rovibrational excitation, enabling reliable application at elevated temperatures.

The implementation was validated through comparison with published energy levels

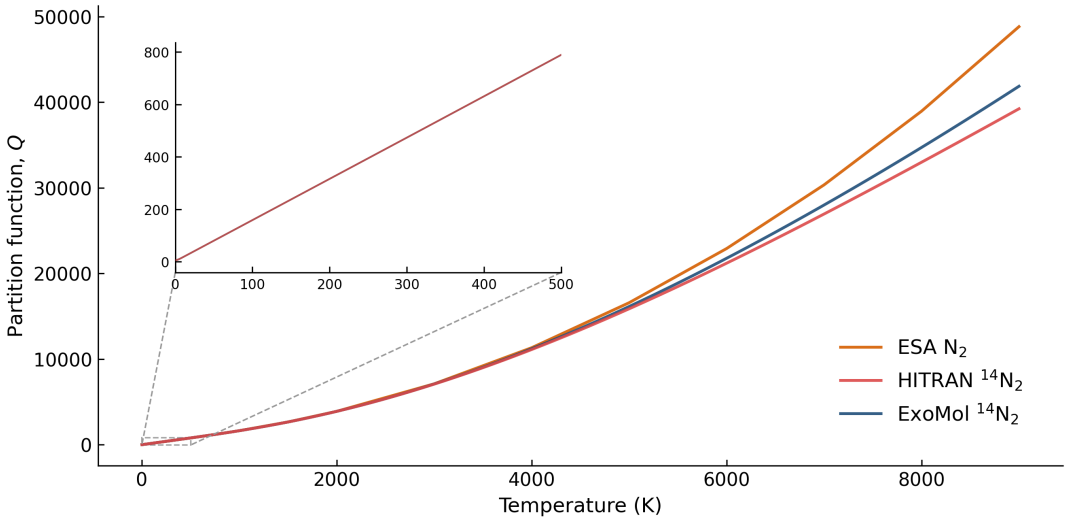

Figure 13: Comparison of the $^{14}N_2$ partition function, Q, from ESA, HITRAN, and ExoMol over 1–9000 K; the ESA values have been scaled as described in the text. The inset shows the low-temperature region from 1 to 500 K.

and existing HITRAN line lists. All 1071 HITRAN transitions for $^{14}N_2$ and 161 transitions for $^{14}N^{15}N$ were recovered. Line positions agree to within $10^{-6}\,cm^{-1}$ for $^{14}N_2$ and show small systematic differences of order $10^{-3}\,cm^{-1}$ for $^{14}N^{15}N$, increasing with J and likely reflecting limitations in the HITRAN reference calculation. Transition intensities were reproduced to sub-percent accuracy, with typical differences of 0.2% for $^{14}N_2$ and better than 0.01% for $^{14}N^{15}N$, not attributable to partition function differences. Transition frequency uncertainties were estimated by extending experimental uncertainties from the fitted dataset to higher rovibrational excitation. Together, these comparisons demonstrate consistent reproduction of the HITRAN line lists within the present implementation.

Calculations were extended to a complete rovibrational basis with $v, J \leq 80$. Transitions with intensities below $10^{-45}\,cm\,molecule^{-1}$ at 296 K were removed as far below any realistic detection threshold, even at elevated temperatures. The remaining line lists were examined for the onset of numerical noise following the approach of Medvedev, identified by a breakdown in the expected monotonic decrease of overtone intensities with increasing Δv. Transitions involving $v \geq 10$ were therefore excluded. The final $^{14}N_2$ line list contains approximately 3700 transitions, extending to $\sim 20000\,cm^{-1}$ with $J_{max} = 69$. Corresponding line lists were produced for $^{14}N^{15}N$ and $^{15}N_2$, the latter not currently available in HITRAN, with slightly reduced coverage due to weaker intensities.

Simulated spectra show that hot-band and high-J transitions contribute significantly

at elevated temperatures, demonstrating the need for extended rovibrational coverage when modelling environments ranging from protoplanetary disks to ultra-hot exoplanet atmospheres. Partition functions computed from the retained energy levels closely match HITRAN at low temperature and show improved convergence at high temperature, lying between HITRAN and another implementation that additionally includes excited electronic states, as expected for a purely rovibrational treatment. Together, these results demonstrate that the present line lists provide a sufficiently complete description of the ground-state spectrum for high-temperature applications, and are provided for all three isotopologues.

Overall, this work has independently reproduced the HITRAN quadrupole line lists for the N_2 ground electronic state and extended isotopologue coverage and rovibrational completeness. The resulting line lists provide transition frequencies with uncertainty estimates and Einstein A coefficients for all three stable isotopologues, including a new line list for $^{15}N_2$, suitable for modelling nitrogen-rich atmospheres across a broad temperature range. Future work could include an analytic representation of the quadrupole moment function to reduce high-overtone numerical noise, filtering based directly on quadrupole matrix elements, and investigation of non-Born–Oppenheimer electric dipole transitions in $^{14}N^{15}N$.

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Appendix

A GitHub repository is provided containing the Duo and ExoCross input and output files, Python scripts, and final line lists in ExoMol format. The included Python code enables straightforward reproduction of tables and comparisons with HITRAN.

https://github.com/jakecb123/ExoMol-MSci-Dissertation